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(54) **ELECTRIC WORK VEHICLE**

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(57) **ABSTRACT**

An electric work vehicle includes a main frame supporting a travel device, and a battery. The electric work vehicle includes a support frame that is attached to the main frame and on which the battery is placed to be supported by the main frame. The electric work vehicle includes a motor operable with power supplied from the battery to drive the travel device, and the motor is suspended by the support frame.

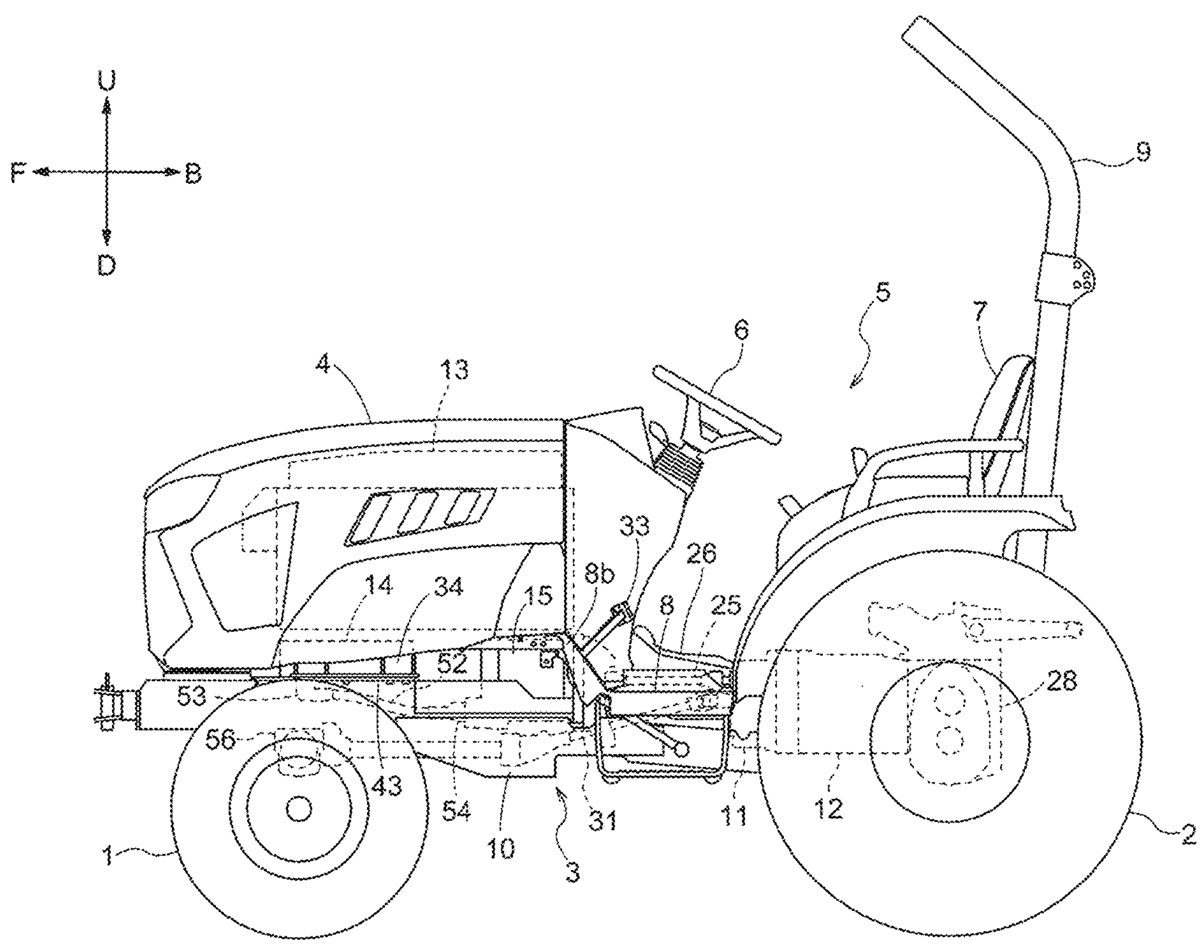


Fig.1

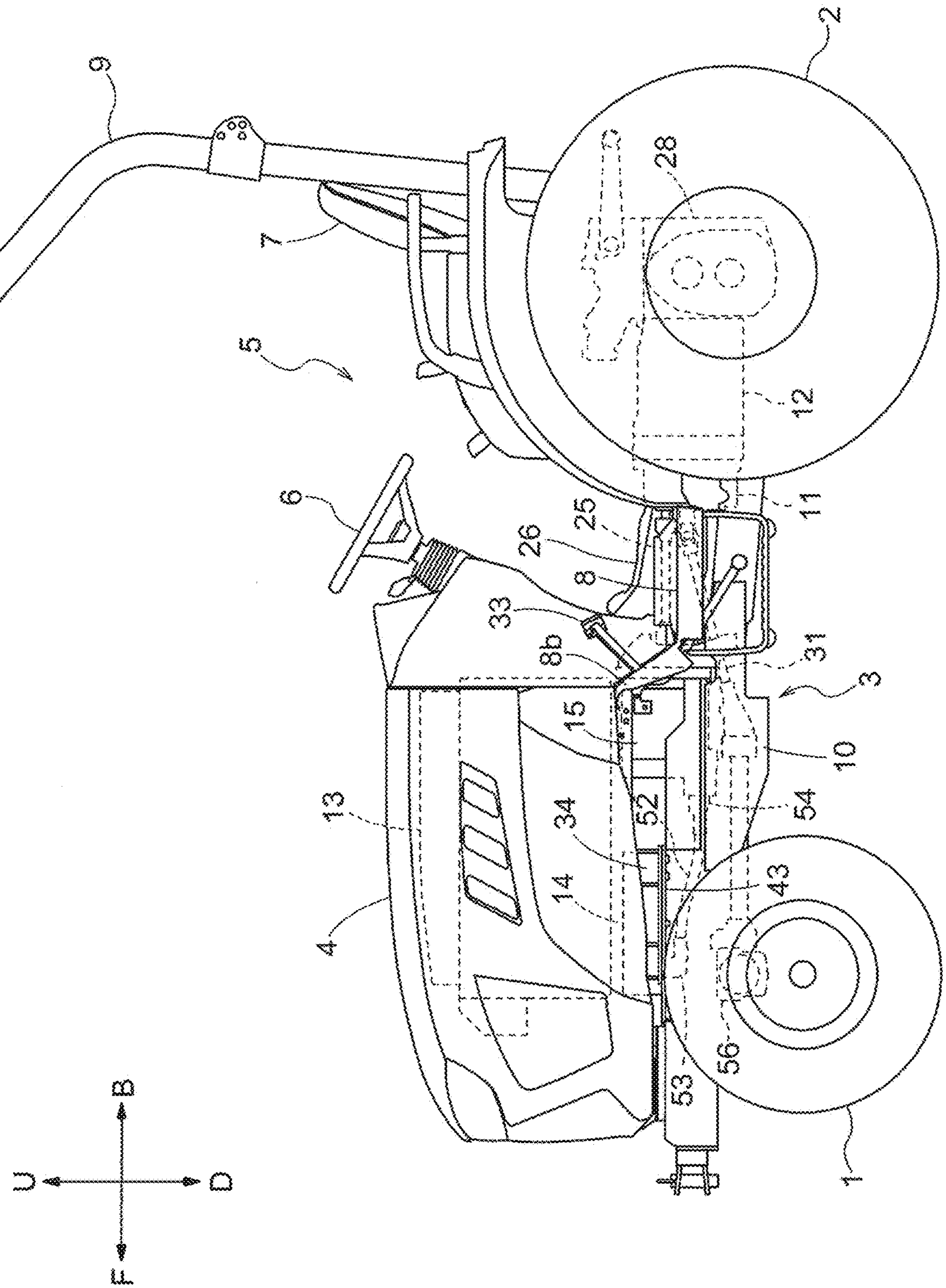


Fig.2

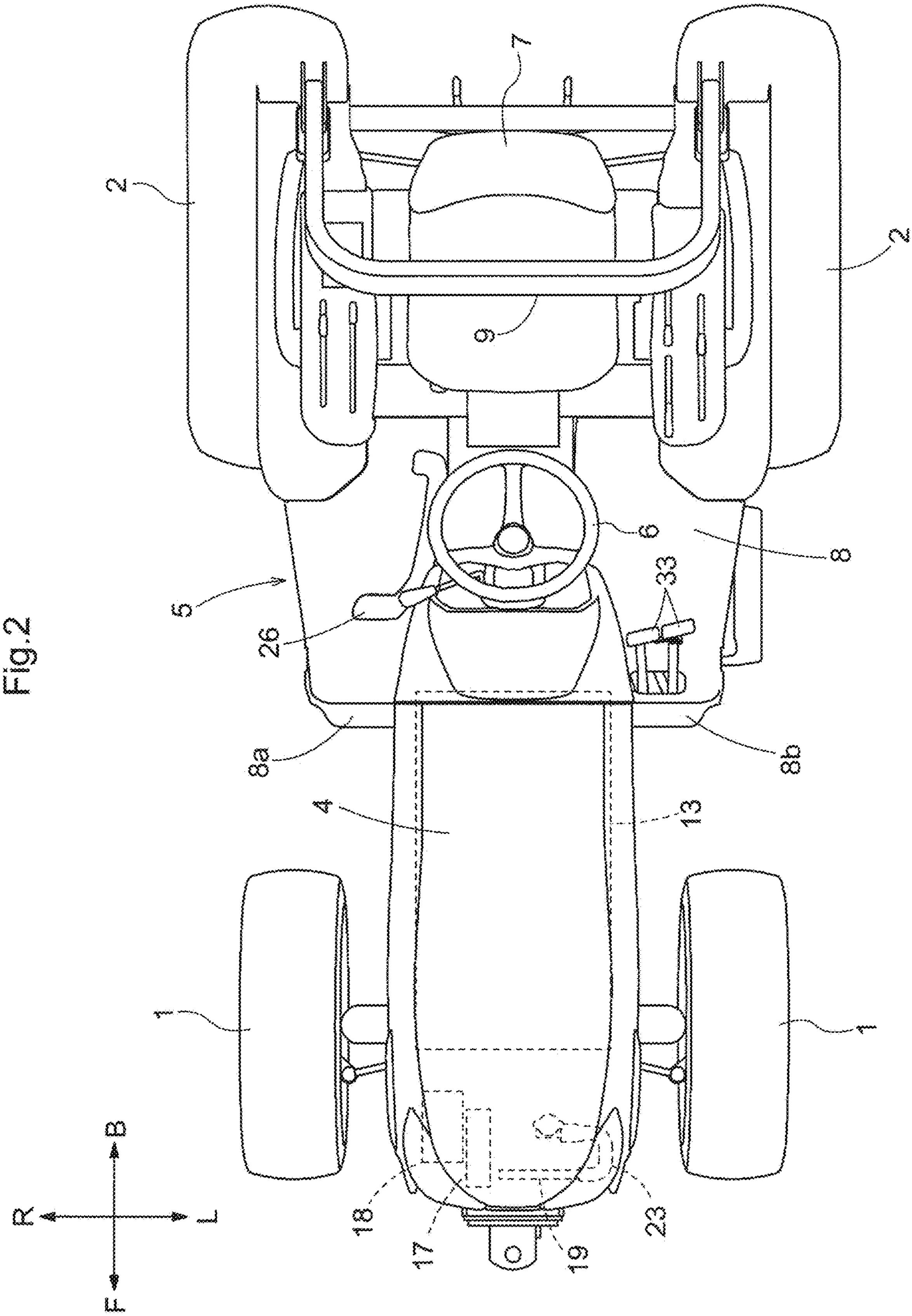
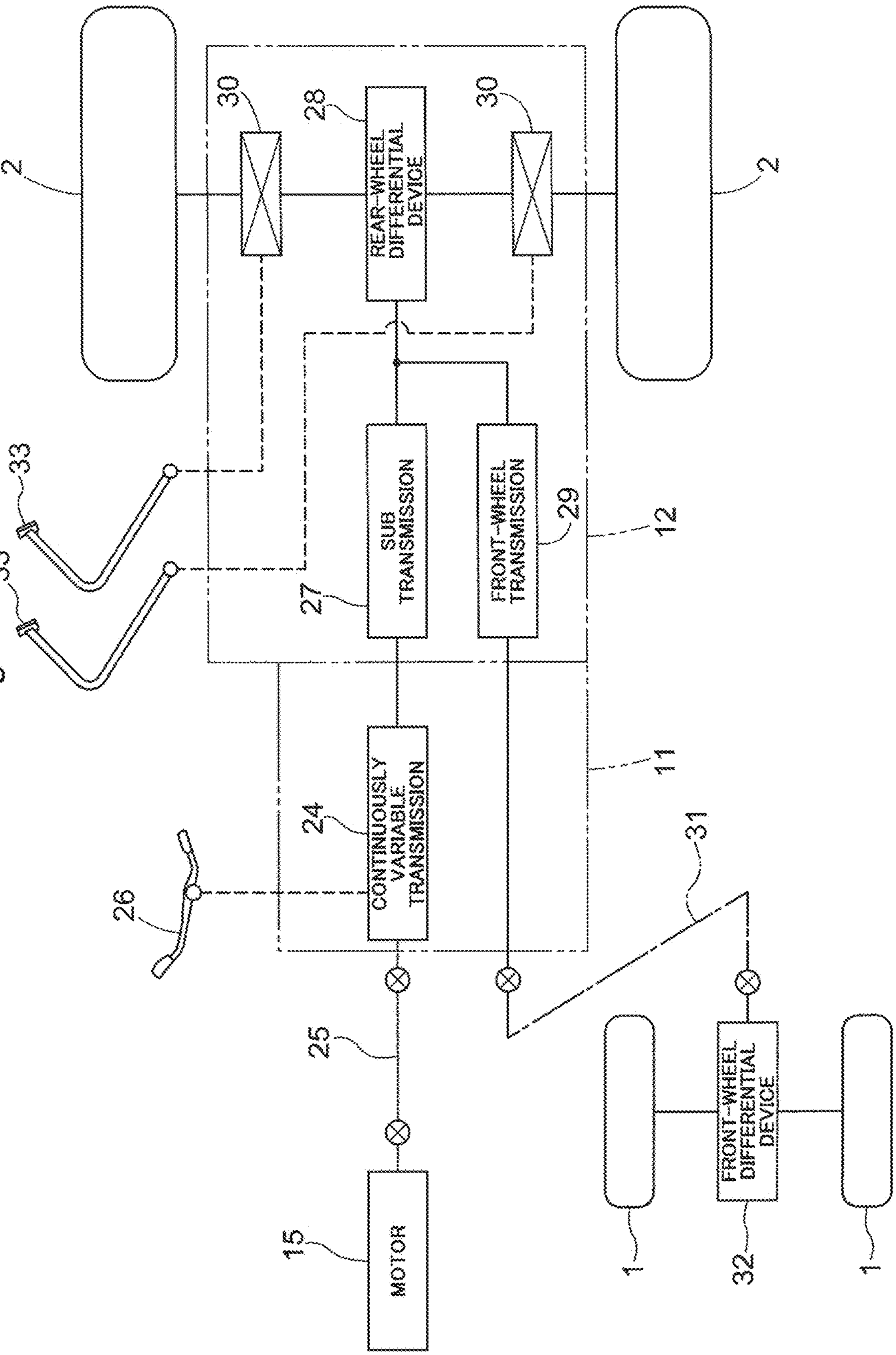


Fig.3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28
REAR-WHEEL DIFFERENTIAL DEVICE
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SUB TRANSMISSION
FRONT-WHEEL TRANSMISSION
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24
CONTINUOUSLY VARIABLE TRANSMISSION
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MOTOR
FRONT-WHEEL DIFFERENTIAL DEVICE
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32
1

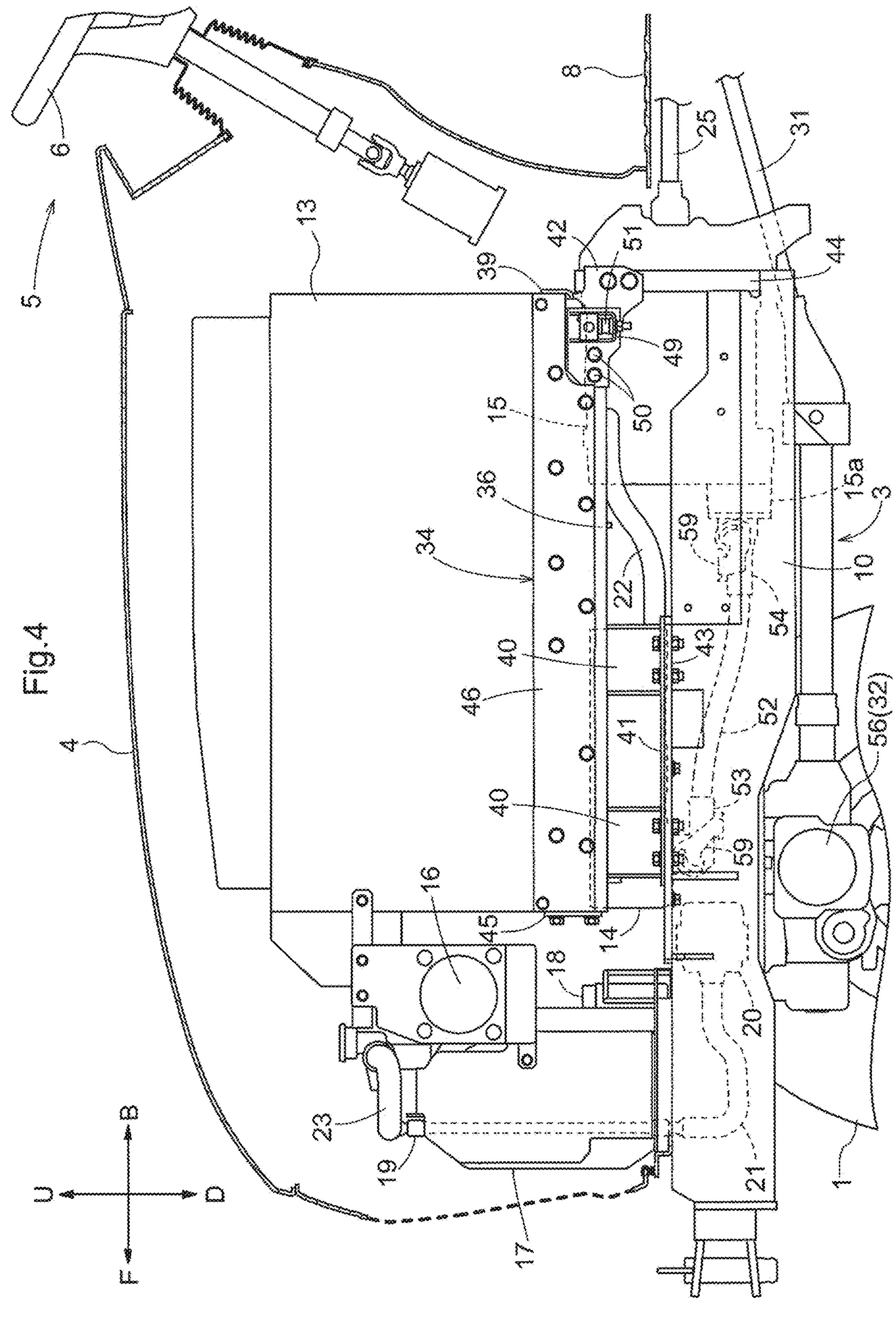
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Fig.4

Fig.5

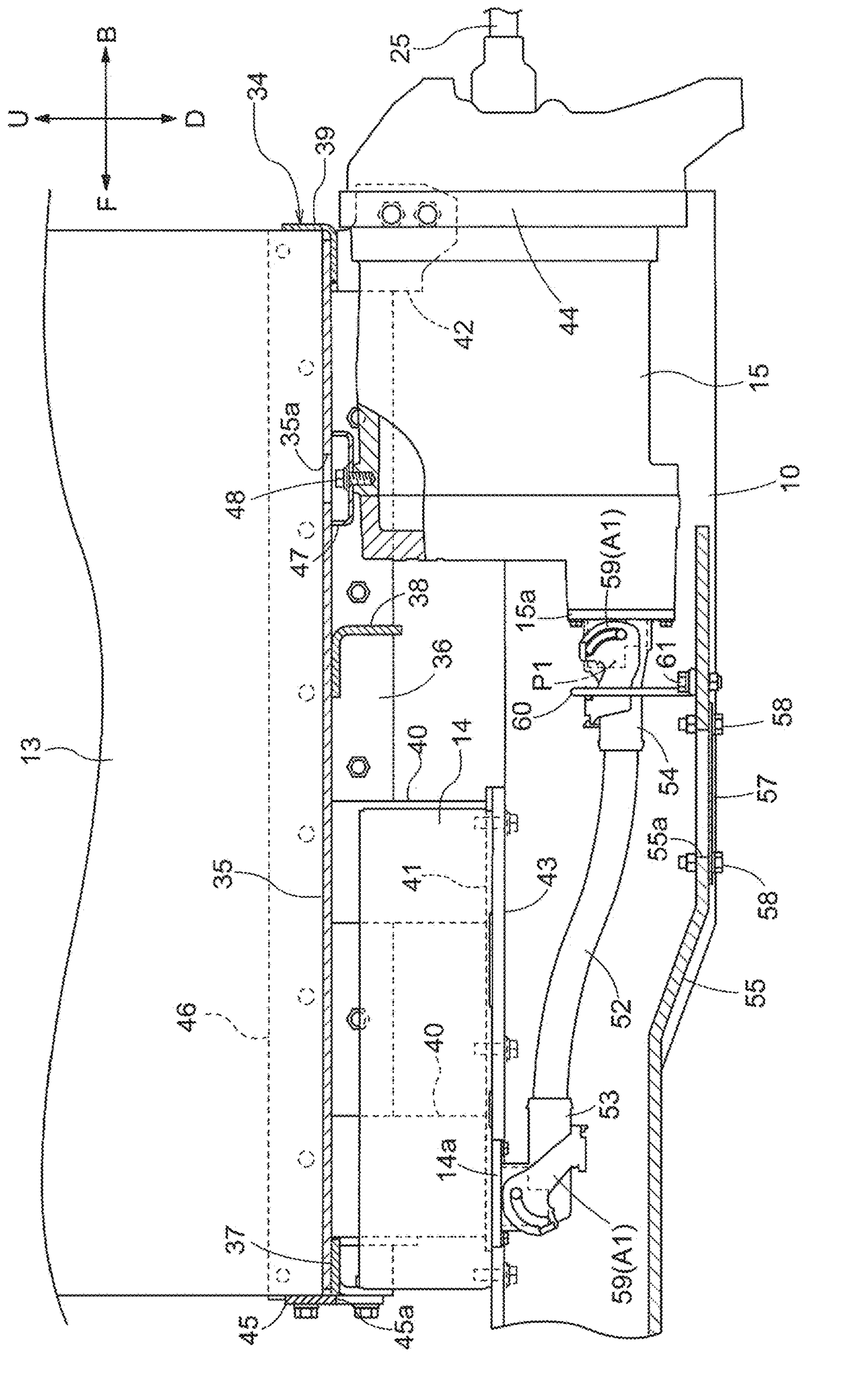
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Fig.6

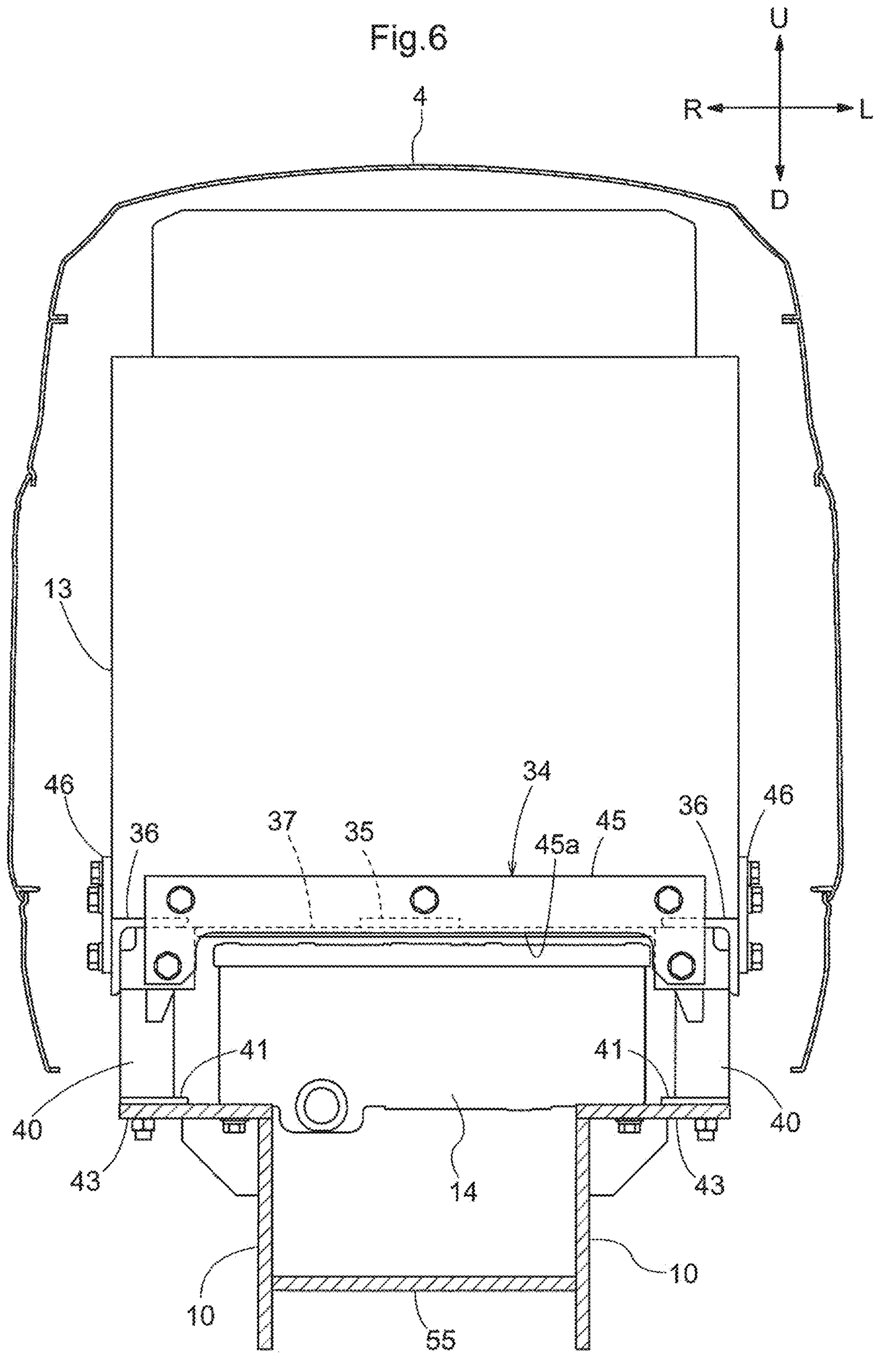

Fig.7

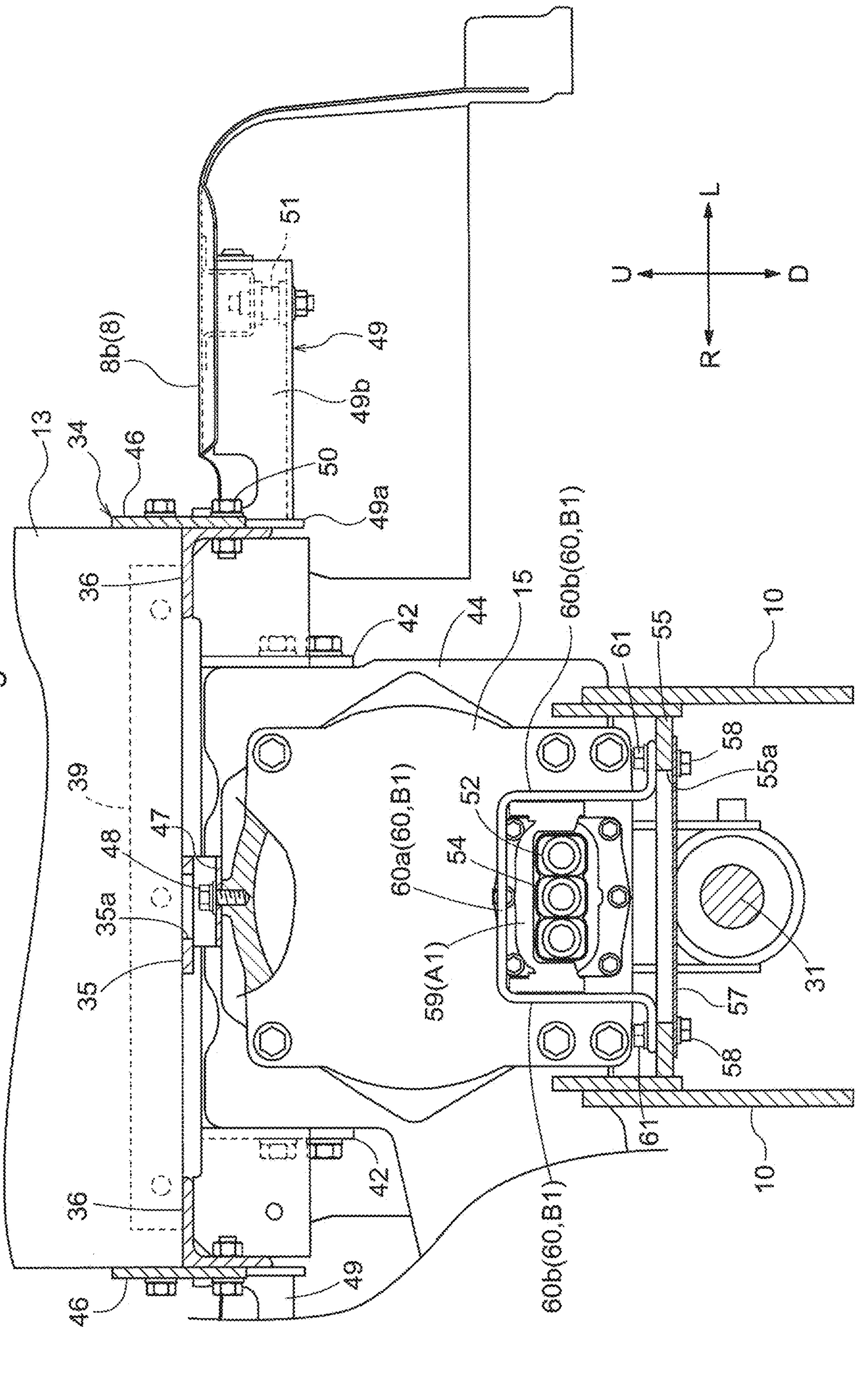

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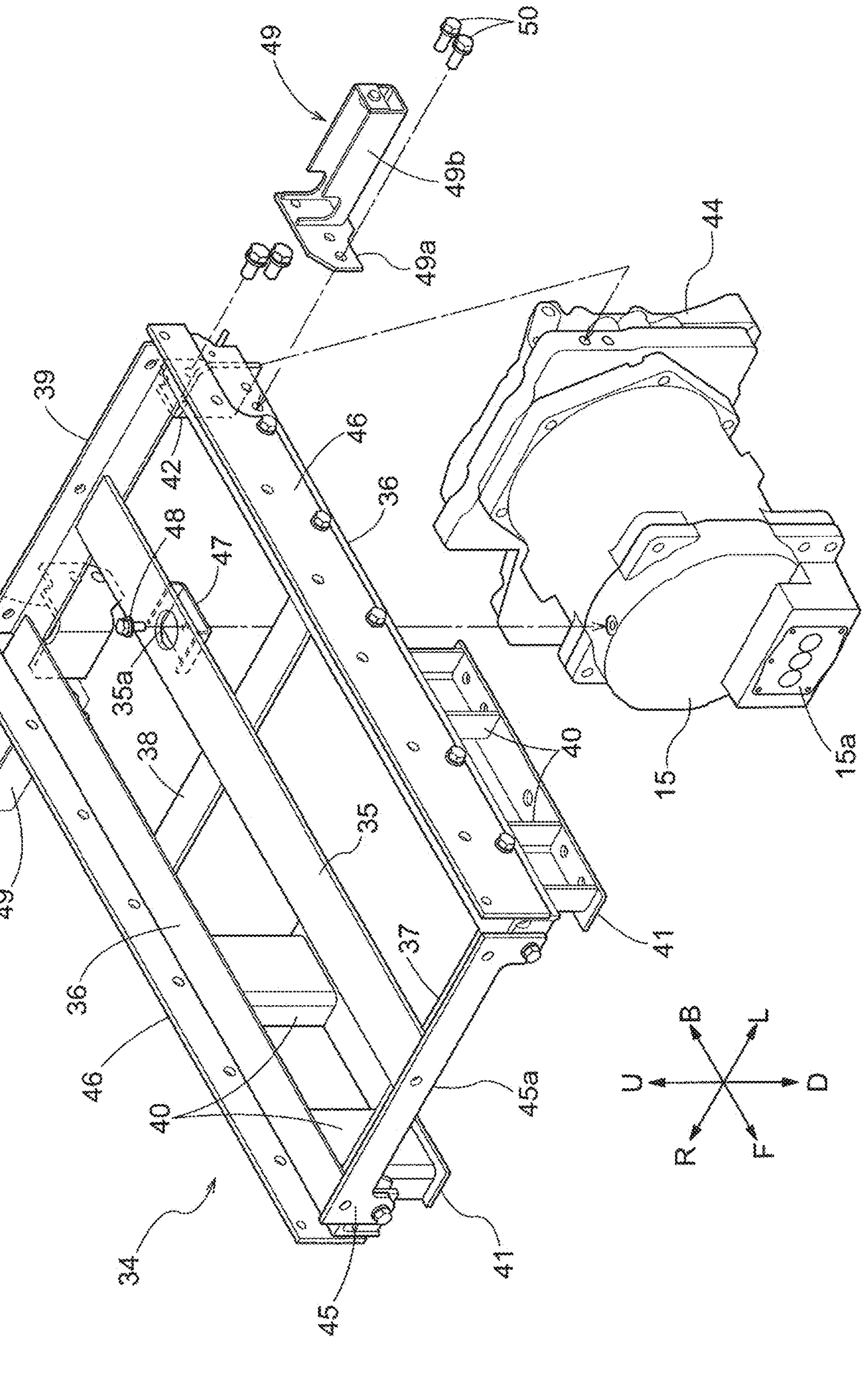
Fig.8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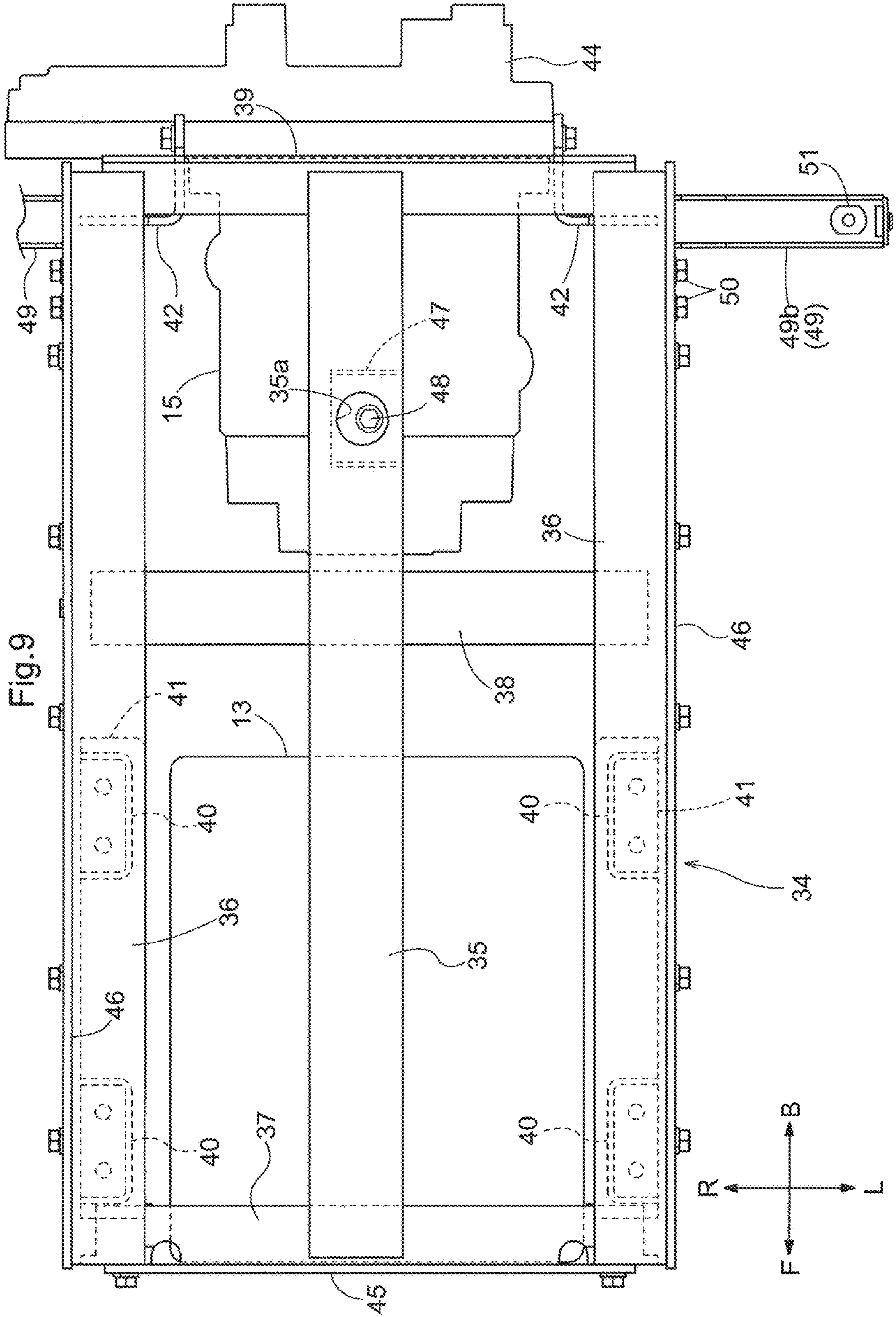
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Fig.9

Fig.10

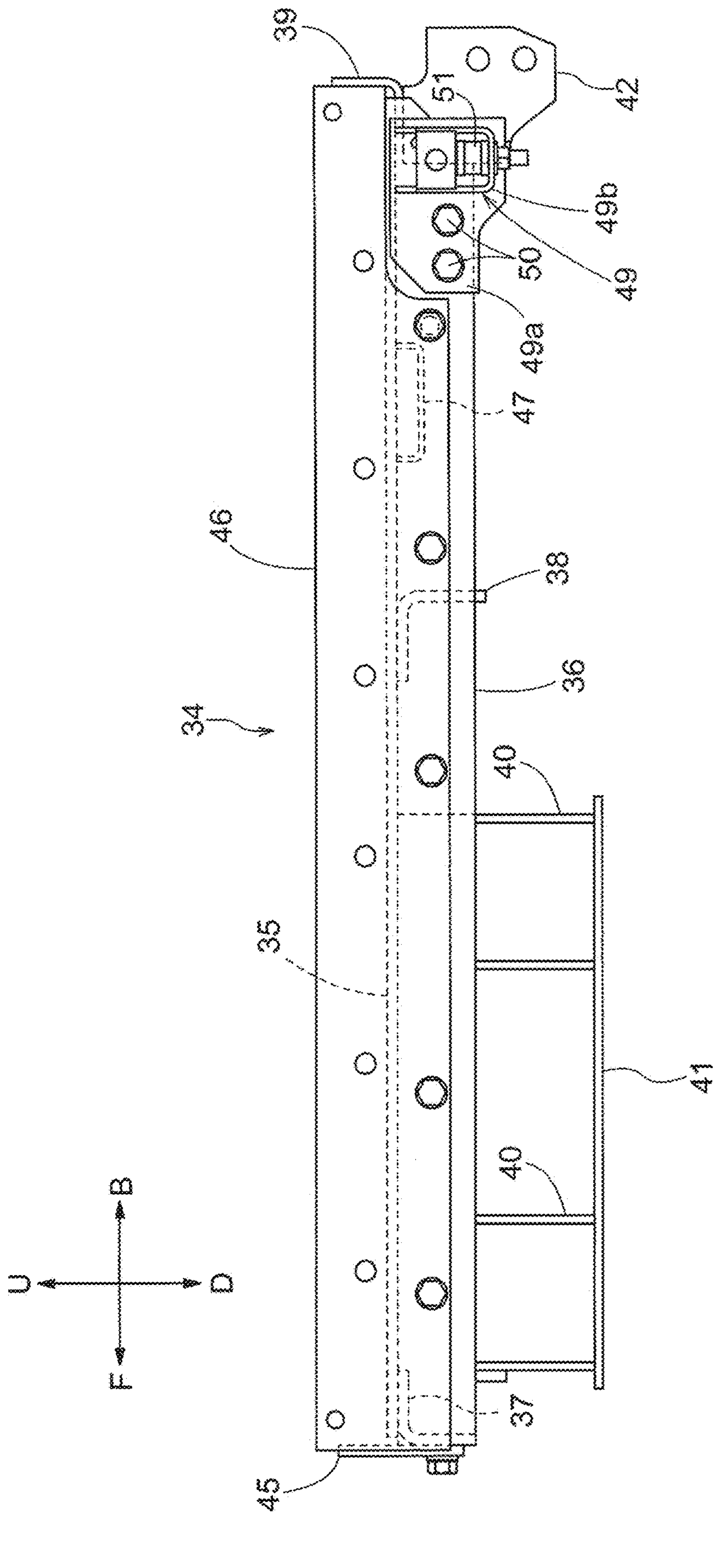

Fig. 11

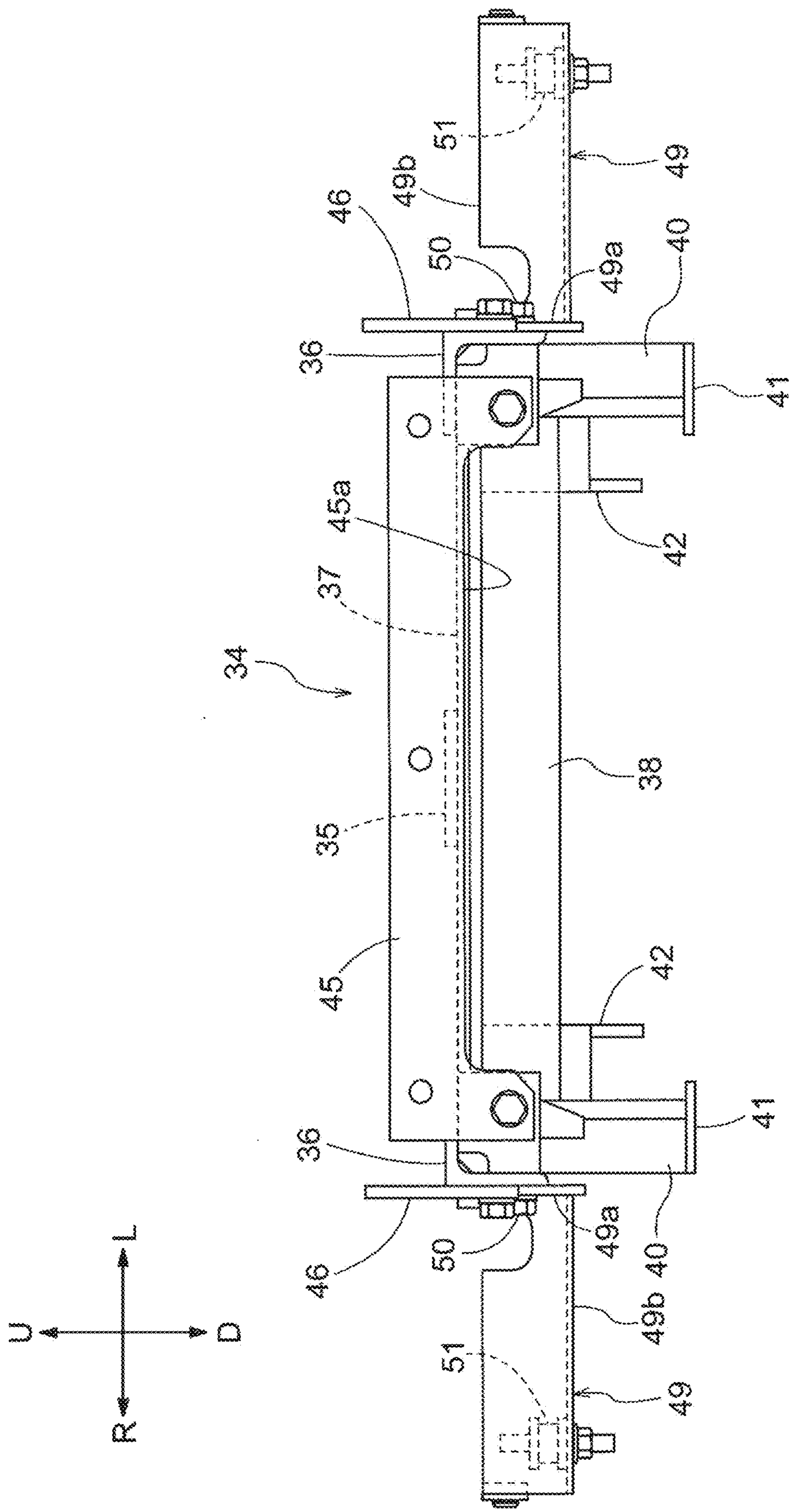

Fig.12

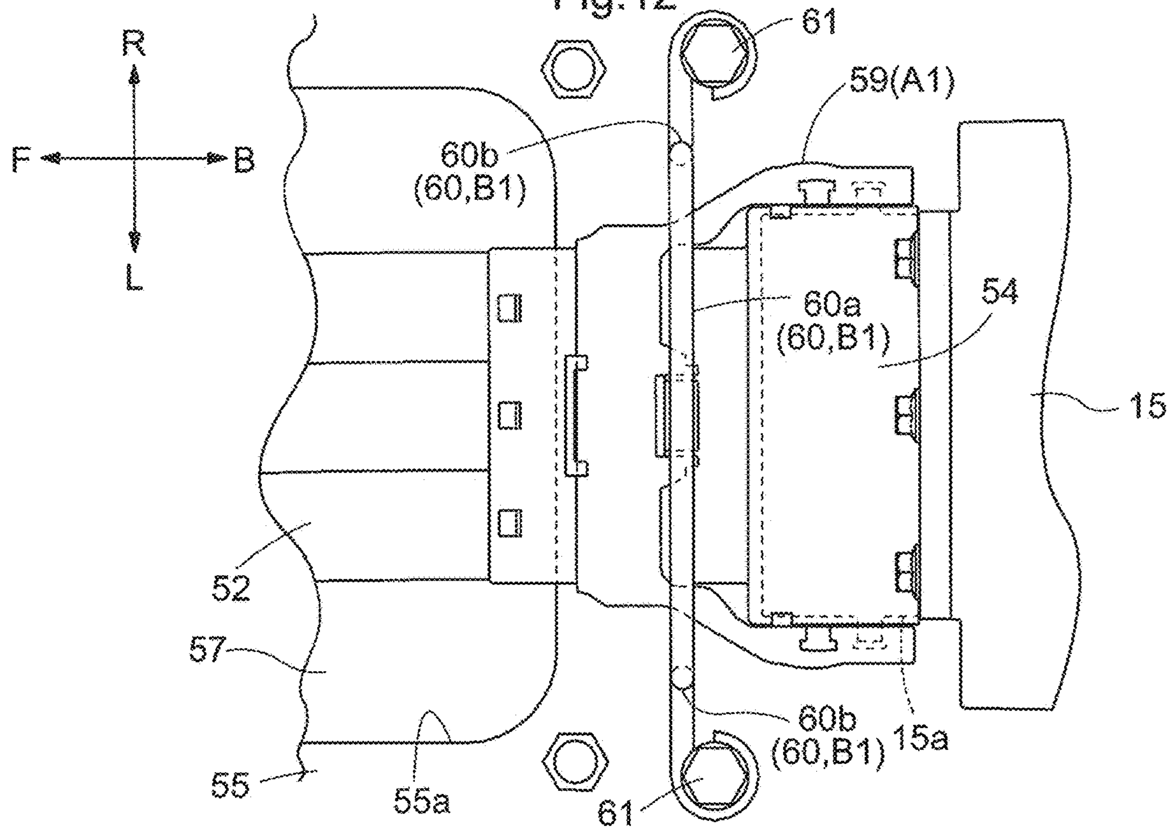
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Fig.13

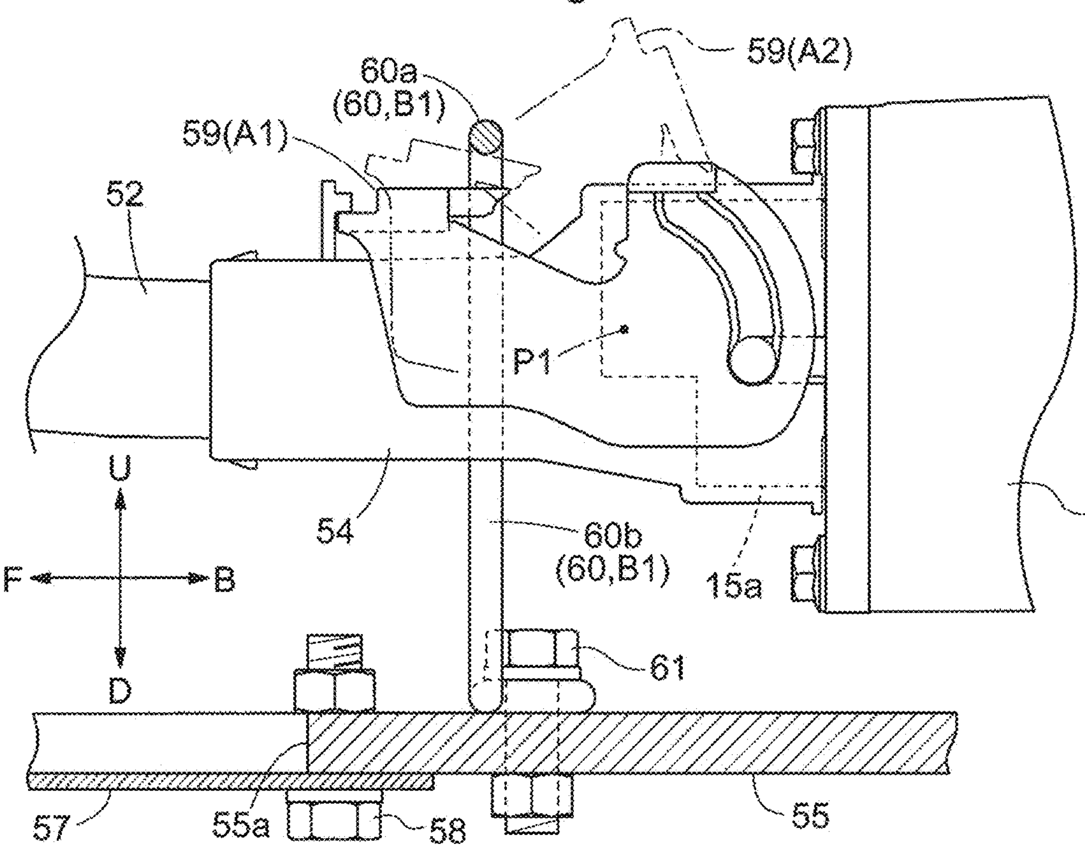
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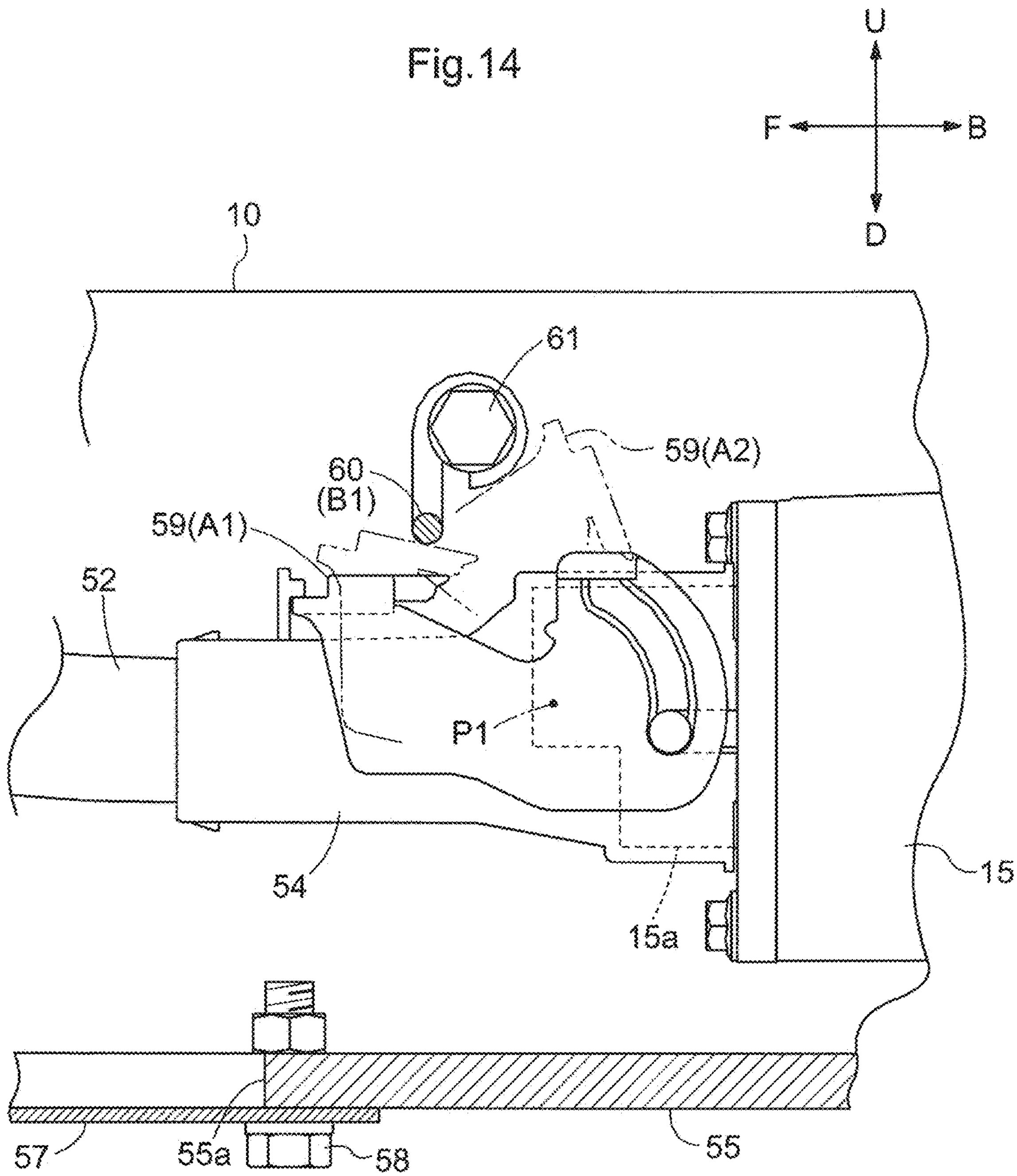
Fig.14



ELECTRIC WORK VEHICLE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2022-210899 filed on Dec. 27, 2022 and is a Continuation Application of PCT Application No. PCT/JP2023/031367 filed on Aug. 30, 2023. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to electric work vehicles each including a travel device configured to be driven by a motor.

2. Description of the Related Art

[0003] Electric work vehicles include an electric work vehicle in which a battery is supported by a main frame and a motor that drives a travel device is provided below the battery and supported by the main frame as disclosed in JP 2020-104568A, for example.

SUMMARY OF THE INVENTION

[0004] In the electric work vehicle disclosed in JP 2020-104568A, various frames are coupled between the motor and the main frame to support the motor with the main frame, and there is room for improvement in simplification of the structure.

[0005] Example embodiments of the present invention simplify structures that support motors in electric work vehicles.

[0006] An electric work vehicle according to an example embodiment of the present invention includes a travel device, a main frame supporting the travel device, a battery, a support frame that is attached to the main frame and on which the battery is placed to be supported by the main frame, and a motor operable with power supplied from the battery to drive the travel device, wherein the motor is suspended by the support frame.

[0007] According to an example embodiment of the present invention, the battery is supported by the main frame via the support frame, and the motor is suspended by the support frame. The motor is supported by the main frame via the support frame, and the support frame supports the battery with the main frame and supports the motor with the main frame.

[0008] According to an example embodiment of the present invention, the support frame has both the function of supporting the battery with the main frame and the function of supporting the motor with the main frame, and accordingly, the structure that supports the motor can be simplified with use of a structural element configured to provide or perform a plurality of functions.

[0009] It is preferable that an electric work vehicle according to an example embodiment of the present invention includes a coupling portion that protrudes downward from the support frame and to which the motor is coupled, and a portion of the support frame above the coupling portion includes an opening that exposes the coupling portion upward.

[0010] According to this example embodiment of the present invention, the coupling portion protrudes downward from the support frame, and accordingly, even if a coupler that couples the motor to the coupling portion is provided on the coupling portion, the coupler does not protrude upward from the support frame. This avoids interference between the coupler and the battery placed on the support frame.

[0011] According to an example embodiment of the present invention, the support frame includes an opening that exposes the coupling portion upward, and therefore, it is easy for a worker to visually check the state of the coupling portion from above through the opening.

[0012] In an example embodiment of the present invention, it is preferable that the support frame includes a plurality of front-rear frames extending along a front-rear direction and spaced apart from each other in a left-right direction and a plurality of left-right frames extending along the left-right direction and spaced apart from each other in the front-rear direction, and the opening is provided in any of the plurality of front-rear frames and the plurality of left-right frames.

[0013] According this example embodiment of the present invention, the support frame has a lattice shape defined by the front-rear frames and the left-right frames, and the worker can see below through gaps between the front-rear frames and gaps between the left-right frames when the battery is removed from the support frame.

[0014] Therefore, the worker can visually check the state of the coupling portion through the gaps between the front-rear frames and the gaps between the left-right frames, as well as being capable of visually checking the state of the coupling portion from above through the opening. Since the support frame has the lattice shape, the weight of the support frame can be reduced.

[0015] In an example embodiment of the present invention, it is preferable that an upper portion of the motor is coupled to the support frame, and a front portion or a rear portion of the motor is coupled to the main frame.

[0016] According to this example embodiment of the present invention, in addition to the upper portion of the motor being coupled to the support frame, the front portion or the rear portion of the motor is coupled to the main frame, and therefore, the motor is supported with higher strength.

[0017] In an example embodiment of the present invention, it is preferable that the support frame includes a base portion on which the battery is placed and an upright portion extending upward from the base portion and surrounding an outer periphery of a lower portion of the battery, an inverter to supply power from the battery to the motor is provided next to the motor under the support frame, and a portion of the support frame adjacent to the inverter has a shape that allows movement of the inverter in a horizontal direction.

[0018] According to an example embodiment of the present invention, the outer periphery of the lower portion of the battery is surrounded by the upright portion of the support frame when the battery is placed on the base portion of the support frame, and therefore, the battery is supported with higher stability when the battery is placed on the support frame.

[0019] According to this example embodiment of the present invention, the inverter configured to supply power from the battery to the motor is provided next to the motor under the support frame.

[0020] In this case, the worker may move the inverter in a horizontal direction to take out the inverter from the position under the support frame and return the inverter to the position under the support frame for maintenance of the inverter.

[0021] According to this example embodiment of the present invention, the portion of the support frame adjacent to the inverter has a shape that allows movement of the inverter in the horizontal direction, and therefore, when the worker moves the inverter in the horizontal direction as described above, the inverter is less likely to come into contact with the portion of the support frame adjacent to the inverter, and this facilitates the maintenance work.

[0022] In an example embodiment of the present invention, it is preferable that the portion of the support frame adjacent to the inverter is recessed upward.

[0023] According to this example embodiment of the present invention, the portion of the support frame adjacent to the inverter has a simple shape that is recessed upward, and thus, it is possible to easily obtain a configuration that makes the inverter less likely to come into contact with the portion of the support frame adjacent to the inverter, and accordingly, it is possible to simplify the structure of the support frame.

[0024] In an example embodiment of the present invention, it is preferable that the portion of the support frame adjacent to the inverter is a portion of the support frame located on a side opposite to the motor with respect to the inverter.

[0025] When the worker removes the inverter as described above, the worker moves the inverter from the position under the support frame in a direction opposite to the motor. When the worker attaches the inverter, the worker moves the inverter from a position opposite to the motor with respect to the support frame to the position under the support frame. Accordingly, the motor is unlikely to obstruct the movement of the inverter, and the maintenance work is facilitated.

[0026] It is preferable that the electric work vehicle according to an example embodiment of the present invention includes a driving section behind the support frame, a right front portion of the driving section extends forward to a position adjacent to a right rear portion of the support frame and a left front portion of the floor section extends forward to a position adjacent to a left rear portion of the support frame, the electric work vehicle further includes right and left floor frames capable of supporting the right front portion and the left front portion of the floor section, and the right and left floor frames are attachable to and detachable from the right rear portion and the left rear portion of the support frame.

[0027] In the electric work vehicle according to this example embodiment of the present invention, the driving section may be provided behind the support frame, and the right front portion and the left front portion of the floor section of the driving section may extend forward to positions adjacent to the right rear portion and the left rear portion of the support frame.

[0028] According to this example embodiment of the present invention, the right front portion and the left front portion of the floor section are supported by the right and left floor frames attached to the right rear portion and the left rear portion of the support frame, and therefore, the floor section is supported with higher stability.

[0029] According to this example embodiment of the present invention, the floor frames are attachable to and detachable from the support frame, and therefore, when the floor section is to be removed or attached for the maintenance work, the floor frames can be removed beforehand. Alternatively, the floor frames can be removed and attached when the floor section is removed and attached.

[0030] This makes it possible to remove and attach the floor frames according to how the work for removing or attaching the floor section is performed and how the maintenance work is performed, and therefore, the work for removing or attaching the floor section and the maintenance work are facilitated and work efficiency is improved.

[0031] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] FIG. **1** is a left side view of a tractor.

[0033] FIG. **2** is a plan view of the tractor.

[0034] FIG. **3** is a schematic diagram showing a transmission system for transmission from a motor to front wheels and rear wheels.

[0035] FIG. **4** is a left side view showing the inside of a hood.

[0036] FIG. **5** is a left side view showing the vicinity of an inverter and the motor.

[0037] FIG. **6** is a front view of a vertical cross section showing the vicinity of a support frame and the inverter.

[0038] FIG. **7** is a front view of a vertical cross section showing the vicinity of the motor.

[0039] FIG. **8** is an exploded perspective view of the support frame and the motor.

[0040] FIG. **9** is a plan view of the support frame.

[0041] FIG. **10** is a left side view of the support frame.

[0042] FIG. **11** is a front view of the support frame.

[0043] FIG. **12** is a plan view showing the vicinity of a terminal of the motor, a connector, and a restriction structure.

[0044] FIG. **13** is a left side view of a vertical cross section showing the vicinity of the terminal of the motor, the connector, and the restriction structure.

[0045] FIG. **14** is a left side view of a vertical cross section showing the vicinity of the terminal of the motor, the connector, and the restriction structure in a variation of an example embodiment of the present invention.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0046] FIGS. **1** to **14** show tractors as examples of electric work vehicles. In FIGS. **1** to **14**, F indicates the forward direction, B indicates the backward direction, U indicates the upward direction, D indicates the downward direction, R indicates the rightward direction, and L indicates the leftward direction.

[0047] As shown in FIGS. **1** and **2**, a body **3** is supported by right and left front wheels **1** (corresponding to a travel device) and right and left rear wheels **2** (corresponding to a travel device). A hood **4** is provided in a front portion of the body **3**, and a driving section **5** is provided in a rear portion of the body **3**. A steering wheel **6** for steering the front wheels **1**, a driver's seat **7**, a floor section **8**, and a ROPS frame **9** are provided in the driving section **5**.

[0048] The body **3** includes right and left main frames **10**, a continuously variable transmission case **11**, a transmission case **12**, and the like. The continuously variable transmission case **11** is coupled to a front portion of the transmission case **12**, and the right and left main frames **10** are coupled to the continuously variable transmission case **11** and the transmission case **12** and extend along the front-rear direction. The front wheels **1** are supported by the main frames **10**, and the rear wheels **2** are supported by the transmission case **12**.

[0049] With this configuration, the tractor includes the front wheels **1**, the rear wheels **2**, and the main frames **10** supporting the front wheels **1**.

[0050] As shown in FIG. **1**, a battery **13**, an inverter **14**, and a motor **15** are supported by the main frames **10**.

[0051] As shown in FIG. **4**, a connecting portion **16** is provided in a front left portion of the battery **13**, and a charger (not shown) of an external power source (not shown) is connected to the connecting portion **16** to charge the battery **13** with power supplied from the external power source via the connecting portion **16**. Power is supplied from the battery **13** to the inverter **14**, and the DC power supplied from the battery **13** is converted to AC power in the inverter **14** and supplied to the motor **15** to drive the motor **15**.

[0052] As shown in FIGS. **2** and **4**, a DC-DC converter **17**, a battery **18**, a radiator **19**, a water pump **20**, and the like are provided on front portions of the main frames **10**. The battery **13**, the inverter **14**, the connecting portion **16**, the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are covered by the hood **4**.

[0053] With this configuration, the tractor includes the battery **13** and the hood housing the battery **13**.

[0054] Power is supplied from the battery **13** to the DC-DC converter **17** and the voltage is reduced to 12 V in the DC-DC converter **17**. The power whose voltage has been reduced to 12 V is supplied to the battery **13**, and then supplied from the battery **13** to the battery **18**. Various devices (not shown) included in the tractor operate with the power with the voltage of 12 V supplied from the battery **18**.

[0055] When the water pump **20** operates, cooling water generated in the radiator **19** is supplied to the water pump **20** via a pipe **21**, and supplied from the water pump **20** to the inverter **14**. The cooling water that has passed through the inverter **14** is supplied to the motor **15** via a pipe **22**, and the cooling water that has passed through the motor **15** is supplied to the DC-DC converter **17**, and returns from the DC-DC converter **17** to the radiator **19** via a pipe **23**.

[0056] As shown in FIGS. **1** and **3**, a hydraulic continuously variable transmission **24** is housed in the continuously variable transmission case **11**, and motive power is transmitted from the motor **15** to the continuously variable transmission **24** via a transmission shaft **25**. The continuously variable transmission **24** is capable of changing the speed of forward and rearward movements in a stepless manner, and is operated via a gearshift pedal **26** (see FIG. **2**) provided on the floor section **8**.

[0057] A sub transmission **27**, a rear-wheel differential device **28**, a front-wheel transmission **29**, and right and left brakes **30** are housed in the transmission case **12**. Motive power is transmitted from the continuously variable transmission **24** to the sub transmission **27**, and transmitted from the sub transmission **27** to the rear wheels **2** via the rear-wheel differential device **28**.

[0058] Motive power diverging between the sub transmission **27** and the rear-wheel differential device **28** is transmitted to the front-wheel transmission **29**, transmitted from the front-wheel transmission **29** to a front-wheel differential device **32** via a transmission shaft **31**, and then transmitted from the front-wheel differential device **32** to the front wheels **1**.

[0059] When the front wheels **1** are operated to be within the ranges of right and left set angles from a straight travel position, the front wheels **1** are driven by the front-wheel transmission **29** at the same speed as the rear wheels **2**. When the front wheels **1** are steered rightward or leftward past the right or left set angle, the front wheels **1** are driven by the front-wheel transmission **29** at a speed higher than the speed of the rear wheels **2**.

[0060] With this example configuration, the tractor includes the motor **15** that operates with power supplied from the battery **13** and drives the front wheels **1** and the rear wheels **2**, and the front wheels **1** and the rear wheels **2** (travel device) driven by the motor **15**.

[0061] The right brake **30** is provided on a transmission system for transmission from the rear-wheel differential device **28** to the right rear wheel **2**, and the left brake **30** is provided on a transmission system for transmission from the rear-wheel differential device **28** to the left rear wheel **2**. A right brake pedal **33** that can operate the right brake **30** and a left brake pedal **33** that can operate the left brake **30** are provided in a left front portion ***8b*** of the floor section **8** (see FIG. **2**).

[0062] It is possible to make a turn with a small turning radius by stepping on the right (left) brake pedal **33** corresponding to the center of turning when making a right turn (left turn) to apply the right (left) brake **30**.

[0063] As shown in FIGS. **4** to **7**, a support frame **34** is attached to the main frames **10**, and the battery **13** is placed on and coupled to the support frame **34**.

[0064] As shown in FIGS. **8** to **11**, the support frame **34** includes front-rear frames **35** and **36** (corresponding to base portions), left-right frames **37**, **38**, and **39** (corresponding to base portions), right and left leg portions **40**, right and left coupling plates **41**, right and left brackets **42**, upright portions **45** and **46**, and the like.

[0065] The front-rear frames **35** and **36** are angled bars extending along the front-rear direction. The left-right frame **37** is a flat plate extending along the left-right direction. The left-right frames **38** and **39** are angled bars extending along the left-right direction. The front-rear frame **35** and the right front-rear frame **36** are spaced apart from each other in the left-right direction. Also, the front-rear frame **35** and the left front-rear frame **36** are spaced apart from each other in the left-right direction. Accordingly, the front-rear frame **35**, the right front-rear frame **36**, and the left front-rear frame **36** are spaced apart from each other in the left-right direction. The left-right frame **37**, the left-right frame **38**, and the left-right frame **39** are spaced apart from each other in the front-rear direction. The front-rear frames **35** and **36** are coupled to the left-right frames **37**, **38**, and **39**. Accordingly, the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) has a lattice shape in a plan view.

[0066] Two channel-shaped leg portions **40** are coupled to a front portion of the right front-rear frame **36**. A coupling plate **41** extending in the front-rear direction is coupled to lower portions of the two leg portions **40**. Two channel-shaped leg portions **40** are coupled to a front portion of the left front-rear frame **36**. A coupling plate **41** extending in the front-rear direction is coupled to lower portions of the two leg portions **40**. The right bracket **42** is coupled to a rear portion of the right front-rear frame **36**, and the left bracket **42** is coupled to a rear portion of the left front-rear frame **36**.

[0067] The upright portion **45** having a flat plate shape is coupled to the left-right frame **37** with bolts and extends upward from the left-right frame **37**. The right upright portion **46** having a flat plate shape is coupled to the right front-rear frame **36** with bolts. Also, the left upright portion **46** having a flat plate shape is coupled to the left front-rear frame **36** with bolts. Accordingly, the left and right upright portions **46** extend upward from the left and right front-rear frames **36**.

[0068] As shown in FIGS. **4** to **7**, a right support **43** having a flat plate shape is coupled to a front upper portion of the right main frame **10** along the front-rear direction. Also, a left support **43** having a flat plate shape is coupled to a front upper portion of the left main frame **10** along the front-rear direction. A gear case **44** is coupled to the right and left main frames **10** along the left-right direction.

[0069] The right leg portions **40** and the right coupling plate **41** of the support frame **34** are coupled to the right support **43** with bolts, and the left leg portions **40** and the left coupling plate **41** of the support frame **34** are coupled to the left support **43** with bolts. The right bracket **42** of the support frame **34** is coupled to a right portion of the gear case **44** with bolts, and the left bracket **42** of the support frame **34** is coupled to a left portion of the gear case **44** with bolts.

[0070] As described above, the support frame **34** is attached to the main frames **10**, and the front-rear frames **36** of the support frame **34** are located above upper ends of the main frames **10** and slightly spaced apart therefrom.

[0071] As shown in FIGS. **4** to **7**, the battery **13** is placed on the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34**, and an outer periphery of a lower portion of the battery **13** is surrounded by the upright portions **45** and **46** of the support frame **34**.

[0072] A lower front portion of the battery **13** is coupled to the upright portion **45** with bolts, a lower right portion and a lower left portion of the battery **13** are coupled to the upright portions **46** with bolts, and a lower rear portion of the battery **13** is coupled to the left-right frame **39** with bolts to couple the battery **13** to the support frame **34**.

[0073] With this configuration, the tractor includes the support frame **34** that is attached to the main frames **10** and on which the battery **13** is placed to be supported by the main frames **10**.

[0074] The support frame **34** includes the plurality of front-rear frames **35** and **36** extending along the front-rear direction and spaced apart from each other in the left-right direction and the plurality of left-right frames **37**, **38**, and **39** extending along the left-right direction and spaced apart from each other in the front-rear direction.

[0075] The support frame **34** includes the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** on which the battery is placed and the upright portions **45** and **46** extending upward from the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** and surrounding the outer periphery of the lower portion of the battery **13**.

[0076] As shown in FIGS. **8**, **9**, and **10**, a coupling portion **47** is provided in a rear portion (between the left-right frame **38** and the left-right frame **39**) of the front-rear frame **35** of the support frame **34**.

[0077] The coupling portion **47** is formed by bending a plate into a channel shape, and is provided on the support frame **34** by being coupled to a lower surface of the front-rear frame **35** of the support frame **34** so as to protrude downward from the front-rear frame **35** of the support frame **34**.

[0078] A coupling bolt **48** is inserted into an opening in the coupling portion **47**, and a portion of the front-rear frame **35** of the support frame **34** above the coupling portion **47** includes an opening **35*a***. A worker can fasten the coupling bolt **48** and remove the coupling bolt **48** with a tool such as a wrench by inserting the tool into the opening **35*a*** of the front-rear frame **35**.

[0079] As shown in FIGS. **4**, **5**, and **7** to **10**, the motor **15** is under the rear portion (between the left-right frames **38** and **39**) of the front-rear frame **35** of the support frame **34** and between the right and left main frames **10**. The motor **15** is coupled to the coupling portion **47** with the coupling bolt **48**, and a rear portion of the motor **15** is coupled to the gear case **44** with bolts.

[0080] With this configuration, the coupling portion **47** that protrudes downward and to which the motor **15** is coupled is provided on the support frame **34**, and the motor **15** is suspended by the support frame **34**. A portion of the support frame **34** above the coupling portion **47** includes the opening **35*a*** that exposes the coupling portion **47** upward, and the opening **35*a*** is provided in the front-rear frame **35** of the support frame **34**.

[0081] An upper portion of the motor **15** is coupled to the support frame **34** (coupling portion **47**). The rear portion of the motor **15** is coupled to the main frames **10** via the gear case **44**, and the rear portion of the motor **15** is coupled to the continuously variable transmission case **11** and the transmission case **12** via the gear case **44**.

[0082] The motor **15** is located below a lower end portion of the hood **4** and the support frame **34** (front-rear frames **36**) in a side view (see FIGS. **1** and **4**), and the motor **15** can be seen through a gap between upper end portions of the main frames **10** and the lower end portion of the hood **4** and the support frame **34** (front-rear frames **36**).

[0083] With this configuration, the tractor includes the motor **15** located outside the hood **4**.

[0084] As shown in FIGS. **4**, **5**, and **6**, the inverter **14** is supported by the main frames **10** by being coupled to the right and left supports **43** with bolts, and is located at a position next to the motor **15** in the front-rear direction and under a front portion (between the left-right frames **37** and **38**) of the support frame **34**.

[0085] As shown in FIGS. **8** and **11**, a lower portion **45*a*** of the upright portion **45** (corresponding to a portion of the support frame **34** adjacent to the inverter **14**) is recessed upward, and the upright portion **45** has a gate shape in a front view.

[0086] When the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are removed, the worker can see the inverter **14** from the front side through a region under the lower portion **45*a*** of the upright portion **45** as shown in FIG. **6**.

[0087] In the state where the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like are removed, the worker can remove the inverter **14** from the main frames **10** by removing the bolts coupling the inverter **14** to the supports **43** and moving the inverter **14** forward along the horizontal direction through the region under the lower portion **45***a* of the upright portion **45**.

[0088] After maintenance of the inverter **14** or the like is finished, the worker can place the inverter **14** on the supports **43** by moving the inverter **14** rearward along the horizontal direction from a position in front of the main frames **10** through the region under the lower portion **45***a* of the upright portion **45** and couple the inverter **14** to the supports **43** with bolts.

[0089] With this configuration, the inverter **14** that supplies power from the battery **13** to the motor **15** is provided next to the motor **15** under the support frame **34**.

[0090] The upright portion **45** of the support frame **34** adjacent to the inverter **14** is recessed upward to allow movement of the inverter **14** in the horizontal direction.

[0091] The upright portion **45** of the support frame **34** adjacent to the inverter **14** is a portion of the support frame **34** located on a side opposite to the motor **15** with respect to the inverter **14**.

[0092] As shown in FIGS. **1** and **2**, the driving section **5** and the floor section **8** of the driving section **5** are provided behind the battery **13** and the support frame **34**. A right front portion **8***a* and the left front portion **8***b* of the floor section **8** extend forward and diagonally upward to positions adjacent to right rear portions and left rear portions of the battery **13** and the support frame **34**.

[0093] As shown in FIGS. **8** to **11**, right and left floor frames **49** are provided. Each floor frame **49** includes a base plate portion **49***a* having a flat plate shape and a support portion **49***b* having a channel shape and coupled to the base plate portion.

[0094] The base plate portions **49***a* of the right and left floor frames **49** are coupled to a right rear portion and a left rear portion of the support frame **34** (front-rear frames **36**) with bolts **50**. As shown in FIG. **7**, the right front portion **8***a* and the left front portion **8***b* of the floor section **8** are supported by the support portions **49***b* of the right and left floor frames **49** via anti-vibration structures **51**. The floor frames **49** can be removed from the support frame **34** (front-rear frames **36**) by removing the bolts **50**.

[0095] With this configuration, the tractor includes the right and left floor frames **49** capable of supporting the right front portion **8***a* and the left front portion **8***b* of the floor section **8**, and the right and left floor frames **49** are attachable to and detachable from the right rear portion and the left rear portion of the support frame **34**.

[0096] As shown in FIGS. **4** and **5**, the tractor includes a harness **52**, and a connector **53** is provided at a front end portion of the harness **52**, and the connector **53** is connected to a terminal **14***a* provided in a front lower portion of the inverter **14**. A connector **54** is provided at a rear end portion of the harness **52**, and the connector **54** is connected to a terminal **15***a* provided in a front portion of the motor **15**.

[0097] As shown in FIGS. **5**, **6**, and **7**, the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** are between the right and left main frames **10** so as to overlap the right and left main frames **10** in a side view. A cover **55** having a flat plate shape is coupled to lower portions of the right and left main frames **10** so as to extend between front end portions of the main frames **10** and a position under the terminal **15***a* of the motor **15**.

[0098] As shown in FIGS. **1** and **4**, a front axle case **56** supporting the front wheels **1** is supported by a front portion of the cover **55**, and the transmission shaft **31** (see FIG. **3**) extends forward under the cover **55** and is connected to the front-wheel differential device **32** (see FIG. **3**) housed in the front axle case **56**. The water pump **20** described above is attached to an upper surface of the front portion of the cover **55**.

[0099] As shown in FIGS. **5** and **7**, an opening **55***a* for cleaning and inspection is provided in a portion of the cover **55** between the motor **15** and the front axle case **56** (between the connectors **53** and **54**). A covering **57** having a flat plate shape is provided, and the opening **55***a* of the cover **55** is closed with the covering **57** by attaching the covering **57** to the cover **55** with bolts **58**. It is possible to remove the covering **57** from the cover **55** by removing the bolts **58** to open the opening **55***a* of the cover **55**.

[0100] With this configuration, DC power supplied from the battery **13** is converted to AC power in the inverter **14**, and then supplied from the inverter **14** to the motor **15** via the harness **52**. Thus, the tractor includes the harness **52** to supply power from the battery **13** to the motor **15**. The tractor includes the connector **54** provided at an end portion of the harness **52** and connected to the motor **15**.

[0101] The cover **55** covering the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** is provided under the harness **52** and the connectors **53** and **54**.

[0102] The cover **55** includes the opening **55***a* for cleaning and inspection. The covering **57** is attached to the cover **55** to close the opening **55***a*, and the covering **57** is removable from the cover **55**.

[0103] As shown in FIGS. **12** and **13**, a lock **59** is provided on the connector **54** on the upper side of the connector **54**. As shown in FIGS. **5** and **7**, the worker can see the connector **54** and the lock **59** through the gap between the lower end portion of the hood **4** and the upper end portions of the main frames **10** by looking diagonally downward from positions above right and left sides of the body **3**.

[0104] The lock **59** is supported by the connector **54** so as to be pivotable about an axis P**1** extending along the left-right direction and is operable between a retaining position A**1** at which the lock retains connection between the connector **54** and the terminal **15***a* of the motor **15** and a release position A**2** at which the lock allows the connector **54** to be separated from the terminal **15***a* of the motor **15**.

[0105] The retaining position A**1** is set to a position at which the lock is substantially in contact with an upper portion of the connector **54** and that is spaced apart from the motor **15** further forward than the release position A**2** is. The release position A**2** is set to a position that is spaced apart from the connector **54** farther upward than the retaining position A**1** is and is closer to the motor **15** than the retaining position A**1** is.

[0106] When the worker connects the connector **54** to the terminal **15***a* of the motor **15**, the worker connects the connector **54** to the terminal **15***a* of the motor **15** with the lock **59** operated to the release position A**2**, and then operates the lock **59** to the retaining position A**1**.

[0107] When the worker removes the connector **54** from the terminal **15***a* of the motor **15**, the worker operates the lock **59** from the retaining position A**1** to the release position A**2**, and then moves the connector **54** forward from the terminal **15***a* of the motor **15** to remove the connector.

[0108] As shown in FIGS. **4** and **5**, a lock **59** is provided on the connector **53** in the same manner as the connector **54**, on the lower side of the connector **53**. The state shown in FIG. **5** is a state where the lock **59** on the connector **53** is operated to the retaining position A**1**, and the lock **59** can be operated to the release position A**2** by moving the lock **59** downward and forward.

[0109] With this configuration, the tractor includes the lock **59** operable between the retaining position A**1** at which the lock retains connection between the connector **54** and the motor **15** and the release position A**2** at which the lock allows the connector **54** to be separated from the motor **15**.

[0110] The lock **59** is provided on the upper side of the connector **54**, and the release position A**2** is set to a position spaced apart from the connector **54** further than the retaining position A**1** is.

[0111] As shown in FIGS. **7**, **12**, and **13**, a restriction structure **60** is provided for the connector **54**. The restriction structure **60** includes a first portion **60***a* extending along a horizontal direction and a pair of second portions **60***b* extending in the same direction that is the upward direction or the downward direction from an end portion and another end portion of the first portion **60***a*.

[0112] The state shown in FIGS. **7**, **12**, and **13** is a state where lower portions of the second portions **60***b* of the restriction structure **60** are attached to the cover **55** with bolts **61** (corresponding to fixing structures) and the restriction structure **60** is attached to a restricting position B**1**.

[0113] In the state where the restriction structure **60** is attached to the restricting position B**1**, the cover **55** covering the restriction structure **60** is under the restriction structure **60**, and the restriction structure **60** is located between the right and left main frames **10**, overlapping the right and left main frames **10** in a side view.

[0114] In the state where the restriction structure **60** is attached to the restricting position B**1**, the first portion **60***a* of the restriction structure **60** extends along the left-right direction above the connector **54** and the lock **59** operated to the retaining position A**1** across the entire widths of the connector **54** and the lock **59** operated to the retaining position A**1**. The second portions **60***b* of the restriction structure **60** extend along the up-down direction at right and left positions laterally outward of the connector **54** and the lock **59** operated to the retaining position A**1**.

[0115] In the state where the restriction structure **60** is attached to the restricting position B**1**, the first portion **60***a* of the restriction structure **60** is on an operation path of the lock **59** between the retaining position A**1** and the release position A**2**. Therefore, even if the lock **59** moves from the retaining position A**1** toward the release position A**2**, the lock **59** comes into contact with the restriction structure **60** (the first portion **60***a*) and the movement of the lock **59** is blocked, and accordingly, the lock **59** cannot reach the release position A**2**.

[0116] The worker can remove the restriction structure **60** from the cover **55** (the restricting position B**1**) by removing the bolts **61** with a tool (not shown) such as a wrench. When the restriction structure **60** is removed from the cover **55** (the restricting position B**1**), the worker can operate the lock **59** from the retaining position A**1** to the release position A**2** and remove the connector **54** from the terminal **15***a* of the motor **15**.

[0117] With this configuration, the tractor includes the restriction structure **60** provided on the operation path of the lock **59** between the retaining position A**1** and the release position A**2** to hinder the lock **59** from being operated from the retaining position A**1** to the release position A**2**, and the restriction structure **60** is removable. The restriction structure **60** extends across the entire width of the lock **59** operated to the retaining position A**1**. The bolts **61** (fixing structures) to attach the restriction structure **60** are provided, and the restriction structure **60** can be removed by removing the bolts **61** with a tool.

[0118] The restriction structure **60** includes the first portion **60***a* extending along the horizontal direction and the pair of second portions **60***b* extending in the same direction that is the upward direction or the downward direction from an end portion and another end portion of the first portion **60***a*. The first portion **60***a* is located above the connector **54**, the second portions **60***b* are located laterally outward of the connector **54**, and the lower portions of the second portions **60***b* are attached to the cover **55**.

[0119] A configuration is also possible in which the coupling portion **47** is provided on one of the front-rear frames **36** of the support frame **34** and the opening **35***a* is provided in a portion of the front-rear frame **36** of the support frame **34** above the coupling portion **47**.

[0120] A configuration is also possible in which a plurality of coupling portions **47** are provided on the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34** and openings **35***a* are provided in portions of the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** above the coupling portions **47**.

[0121] With this configuration, the opening **35***a* is provided in any of the plurality of front-rear frames **35** and **36** and the plurality of left-right frames **37**, **38**, and **39**.

[0122] The coupling portion **47** may be omitted.

[0123] In this configuration, the motor **15** can be directly coupled to the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) with the coupling bolt **48**. The motor **15** can be supported by the support frame **34** (the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39**) by being suspended with use of a suspension (not shown).

[0124] A configuration is also possible in which the gear case **44** coupled to the right and left main frames **10** is located in front of the motor **15**, the upper portion of the motor **15** is coupled to the coupling portion **47**, and a front portion of the motor **15** is coupled to the main frames **10** via the gear case **44**.

[0125] A configuration is also possible in which the positional relationship between the inverter **14** and the motor **15** is reversed, the motor **15** is located under the front portion of the support frame **34**, and the inverter **14** is located under the rear portion of the support frame **34**.

[0126] The upright portions **45** and **46** of the support frame **34** may be omitted.

[0127] In this configuration, the battery **13** can be placed on the support frame **34** and coupled to the front-rear frames **35** and **36** and the left-right frames **37**, **38**, and **39** of the support frame **34** with bolts.

[0128] A configuration is also possible in which lower portions of the upright portions **46** and lower portions of the front-rear frames **36** of the support frame **34** are recessed upward.

[0129] In this configuration, the upright portions **46** and the front-rear frames **36** of the support frame **34** are portions of the support frame **34** adjacent to the inverter **14**.

[0130] It is possible to remove the inverter **14** from the main frames **10** by removing the right or left leg portions **40**

and the coupling plate **41** of the support frame **34** without removing the DC-DC converter **17**, the battery **18**, the radiator **19**, and the like, and by moving the inverter **14** rightward or leftward along the horizontal direction through a region under the lower portion of the upright portion **46** and the lower portion of the front-rear frame **36**.

[0131] As shown in FIG. **14**, the restriction structure **60** may also have a straight bar shape.

[0132] In this configuration, the restriction structure **60** is attached to the restricting position B**1** by attaching a right end portion and a left end portion of the restriction structure **60** to the right and left main frames **10** with the bolts **61**.

[0133] With this configuration, the harness **52**, the connectors **53** and **54**, the terminal **14***a* of the inverter **14**, and the terminal **15***a* of the motor **15** are between the right and left main frames **10** so as to overlap the right and left main frames **10** in a side view, and the restriction structure **60** is attached so as to extend between the left and right main frames **10**.

[0134] The restriction structure **60** shown in FIGS. **7**, **12**, and **13** and the restriction structure **60** of a variation of an example embodiment of the present invention may also be configured as described below, rather than being completely removed from the cover **55** (the main frames **10**).

[0135] The restriction structure **60** is configured such that the lock **59** can be operated from the retaining position A**1** to the release position A**2** when an end portion (one second portion **60***b*) of the restriction structure **60** is removed and the position of the restriction structure **60** is changed with the other end portion (the other second portion **60***b*) of the restriction structure **60** kept attached to the cover **55** (the main frames **10**).

[0136] In this configuration, the restriction structure **60** is removed from the restricting position B**1** by changing the position of the restriction structure **60** from the restricting position B**1**. Since the other end portion (the other second portion **60***b*) of the restriction structure **60** is kept attached to the cover **55** (the main frames **10**), the restriction structure **60** will not be lost.

[0137] A configuration is also possible in which the positional relationship between the retaining position A**1** and the release position A**2** of the lock **59** shown in FIGS. **12** and **13** is reversed, and the release position A**2** is set to a position at which the lock is substantially in contact with the upper portion of the connector **54**, and the retaining position A**1** is set to a position that is spaced apart from the connector **54** further upward than the release position A**2** is.

[0138] In this configuration, the restriction structure **60** can be located between the connector **54** and the lock **59** operated to the retaining position A**1** when the restriction structure **60** is attached to the restricting position B**1**.

[0139] In this case, even if the lock **59** moves from the retaining position A**1** toward the release position A**2** to be closer to the connector **54**, the lock **59** comes into contact with the restriction structure **60** and the movement of the lock **59** is blocked, and accordingly, the lock **59** cannot reach the release position A**2**.

[0140] The lock **59** shown in FIGS. **12** and **13** may be provided on the lower side of the connector **54**. The lock **59** may be provided on the lower side of the connector **54**.

[0141] In this configuration, the restriction structure **60** can be provided on the lower side of the connector **54** and the lock **59** as shown in FIG. **14**.

[0142] The lock **59** shown in FIGS. **12** and **13** may also be provided on the motor **15**. The lock **59** may also be provided on the motor **15**.

[0143] In this configuration, when the connector **54** is removed from the terminal **15***a* of the motor **15**, the lock **59** remains on the motor **15**.

[0144] The lock **59** shown in FIGS. **12** and **13** may also be configured to be slid along a horizontal direction or the up-down direction between the retaining position A**1** and the release position A**2**. The lock **59** may also be configured to be slid along a horizontal direction or the up-down direction between the retaining position A**1** and the release position A**2**.

[0145] The configuration of the lock **59** shown in FIGS. **12** and **13** may also be applied to the lock **59** of the connector **53** (see FIG. **5**) connected to the terminal **14***a* of the inverter **14**. The lock **59** of the above-described variations of example embodiments of the present invention may also be applied to the lock **59** of the connector **53** (see FIG. **5**) connected to the terminal **14***a* of the inverter **14**.

[0146] The tractor may include a crawler device (not shown) as a travel device instead of the rear wheels **2**.

[0147] The tractor may include crawler devices (not shown) as travel devices instead of the front wheels **1** and the rear wheels **2**.

[0148] Example embodiments of the present invention are applicable not only to tractors but also to other electric work vehicles such as electric carts, transport vehicles, and the like.

[0149] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

What is claimed is:

1. An electric work vehicle comprising:

a travel device;

a main frame supporting the travel device;

a battery;

a support frame that is attached to the main frame and on which the battery is placed to be supported by the main frame; and

a motor operable with power supplied from the battery to drive the travel device; wherein

the motor is suspended by the support frame.

2. The electric work vehicle according to claim **1**, further comprising:

a coupling portion that protrudes downward from the support frame and to which the motor is coupled; wherein

a portion of the support frame above the coupling portion includes an opening that exposes the coupling portion upward.

3. The electric work vehicle according to claim **2**, wherein the support frame includes:

a plurality of front-rear frames extending along a front-rear direction and spaced apart from each other in a left-right direction; and

a plurality of left-right frames extending along the left-right direction and spaced apart from each other in the front-rear direction; and

the opening is provided in any of the plurality of front-rear frames and the plurality of left-right frames.

4. The electric work vehicle according to claim **1**, wherein an upper portion of the motor is coupled to the support frame, and a front portion or a rear portion of the motor is coupled to the main frame.

5. The electric work vehicle according to claim **1**, wherein the support frame includes:

a base portion on which the battery is placed; and

an upright portion extending upward from the base portion and surrounding an outer periphery of a lower portion of the battery;

an inverter to supply power from the battery to the motor is provided next to the motor under the support frame; and

a portion of the support frame adjacent to the inverter has a shape that allows movement of the inverter in a horizontal direction.

6. The electric work vehicle according to claim **5**, wherein the portion of the support frame adjacent to the inverter is recessed upward.

7. The electric work vehicle according to claim **5**, wherein the portion of the support frame adjacent to the inverter is a portion of the support frame located on a side opposite to the motor with respect to the inverter.

8. The electric work vehicle according to claim **1**, further comprising:

a driving section behind the support frame; wherein

a right front portion of the driving section extends forward to a position adjacent to a right rear portion of the support frame and a left front portion of the floor section extends forward to a position adjacent to a left rear portion of the support frame;

the electric work vehicle further includes right and left floor frames capable of supporting the right front portion and the left front portion of the floor section; and

the right and left floor frames are attachable to and detachable from the right rear portion and the left rear portion of the support frame.

* * * * *